

Scalable Data Science
NPTEL 2026
Assignment-2

1. A hash table stores n elements in m buckets using separate chaining to resolve collisions. Which of the following represents the worst-case time complexity of searching for an element y ?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(1 + l(y))$, where $l(y)$ is the number of elements in the bucket containing y
 - (d) $O(m)$

Solution: Answer:(b) In the worst case, all n elements hash to the same bucket, so searching requires scanning all elements; hence the worst-case time complexity is $O(n)$.

2. A hash table currently has a load factor of 1.2. Which of the following collision-resolution schemes could have been used to construct this table?
 - (a) Open addressing only
 - (b) Chaining only
 - (c) Either open addressing or chaining
 - (d) Neither open addressing nor chaining

Solution: Answer:(b) Since a load factor greater than 1 is possible only with separate chaining.

3. A hash family H is strongly k -wise universal if for any k distinct keys $(x_1, \dots, x_k)$ and any k bucket indices $(y_1, \dots, y_k)$,

$$P(h(x_1) = y_1 \wedge \dots \wedge h(x_k) = y_k) = \frac{1}{m^k}.$$

To construct such a family over $\mathbb{Z}_p$, we use a random polynomial:

$$f(x) = \sum_{i=0}^{k-1} a_i x^i \pmod{p}.$$

What is the optimal storage cost of describing one hash function from this family?

- (a) A. $\mathcal{O}(m^k \log p)$
- (b) B. $\mathcal{O}(k \log p)$
- (c) C. $\mathcal{O}(p^k \log m)$
- (d) D. $\mathcal{O}(\log k + \log m)$

Solution Answer:(B) A degree $(k - 1)$ polynomial over $\mathbb{Z}_p$ is fully determined by its coefficients:

$$a_0, a_1, \dots, a_{k-1}$$

There are k coefficients, and each lies in $\mathbb{Z}_p$, requiring $\log p$ bits.

Hence total storage:

$$k \cdot \log p = \mathcal{O}(k \log p)$$

4. A hash table with p buckets uses chaining and receives k keys, each hashed independently and uniformly. If X denotes the number of buckets whose linked lists remain empty after all insertions, which of the following is equal to $\mathbb{E}[X]$?

- A. $p \left(1 - \frac{1}{p}\right)^k$
- B. $p \left(1 - \frac{1}{k}\right)^p$
- C. $p \left(1 - \frac{1}{p}\right)^p$
- D. $k \left(1 - \frac{1}{p}\right)^k$

Solution Answer:(A) Each bucket remains empty with probability $\left(1 - \frac{1}{p}\right)^k$, so by linearity of expectation,

$$\mathbb{E}[X] = p \left(1 - \frac{1}{p}\right)^k$$

5. Suppose a Misra-Gries sketch with k counters processes a stream of length m . An element x has true frequency $f_x = \alpha m$, where $\alpha > \frac{1}{k+1}$.

An adversary arranges the stream to maximize underestimation of x .

What is the worst-case relative error?

- (a) A. $\frac{1}{\alpha k}$
- (b) B. $1 - \frac{1}{\alpha k}$
- (c) C. $\frac{1}{\alpha(k+1)}$
- (d) D. 0

Solution Answer:(C) In Misra-Gries, the total number of global decrement operations is at most:

$$\frac{m}{k+1}$$

Each decrement can reduce the effective count of x , so:

$$f_x - \hat{f}_x \leq \frac{m}{k+1}$$

Now divide by $f_x = \alpha m$:

$$\frac{f_x - \hat{f}_x}{f_x} \leq \frac{\frac{m}{k+1}}{\alpha m} = \frac{1}{\alpha(k+1)}$$

6. A Bloom filter is configured optimally so that its false positive rate is bounded by $\lambda = 0.01$. If m denotes the number of bits and n the number of stored elements, which of the following asymptotic relationships between m and n is correct?

- A. $m = 9.6n$
- B. $m = 7n$
- C. $m = 2n$
- D. $m = n \log_2 \left(\frac{e}{\lambda} \right)$

Solution: Answer:(D) For an optimally configured Bloom filter with false positive rate λ , the asymptotically optimal number of bits is

$$m = n \log_2 \left(\frac{e}{\lambda} \right)$$

7. A Bloom filter is used to test whether an element belongs to a set. Which of the following situations can occur due to the probabilistic nature of the Bloom filter?

- A. An inserted element is declared absent.
- B. A non-inserted element is declared present.
- C. Both inserted and non-inserted elements are declared absent.
- D. No incorrect answers are ever produced.

Solution: Answer:(B) A Bloom filter may report a non-inserted element as present (a false positive), but it never reports an inserted element as absent.

8. Lets say we have a Bloom filter of a size 10 and total 4 elements have been inserted. We are using a hash family with 3 hash functions. What is the probability that for a given query element (which has not been inserted yet) the Bloom filter return present?

- A. 0.40
- B. 0.54
- C. 0.10
- D. 0.34

(D)

Probability that lookup operation is giving us present is,

$$Pr[Lookup(B, x) = Present] = (1 - e^{-kn/m})^k$$

where k = number of hash function in hash family, m = size of the bloom filter and n = number of element inserted.

9. Why is selecting a hash function uniformly at random from the family of all functions Y^U (where U is the universe and $Y = \{0, 1, \dots, m-1\}$) considered impractical?

- (a) A. It requires $\mathcal{O}(|U| \log m)$ bits to store the function.
- (b) B. It maps all keys to bucket 0 with probability $1/|U|!$.
- (c) C. It requires $\mathcal{O}(m^2)$ space due to pigeonhole constraints.
- (d) D. It requires $\mathcal{O}(|U|^2)$ time to evaluate.

Solution Answer:(A) A truly random function assigns an independent output in Y to every element in U .

Thus, to describe such a function explicitly, we must store:

$$h(x) \in \{0, 1, \dots, m-1\} \quad \forall x \in U$$

Each output requires $\log m$ bits, and there are $|U|$ inputs.

Hence total storage:

$$|U| \cdot \log m = \mathcal{O}(|U| \log m)$$

This is as large as (or larger than) the dataset itself, making it impractical.

10. You have a Bloom filter with k hash functions, n bits and m inserted elements. If you increase k (the number of hash functions) beyond the value k^* that minimizes the false positive rate, which happens?
- A. False positive rate continues to decrease.
 - B. False positive rate increases beyond k^* .
 - C. The filter becomes useless.
 - D. No effect.

Solution Answer:(B) There is a sweet spot for k ; beyond it, false positives actually become more likely.

11. A data stream is processed using the Flajolet-Martin algorithm for distinct element estimation. If an arriving element maps to a hash value that has already been observed, which of the following is true?
- A. The estimated distinct count increases.
 - B. The estimated distinct count decreases.
 - C. The estimated distinct count remains unchanged.
 - D. A collision error is generated.

Solution: Answer:(C) Repeated hash values do not affect the maximum observed trailing-zero pattern, so the estimated distinct count remains unchanged.

12. For a Bloom filter of size n bits, k hash functions and m inserted items, what is the probability that a given bit is still 0 after all insertions?
- A. $(1 - \frac{1}{n})^{km}$
 - B. $1 - (1 - \frac{1}{n})^{km}$
 - C. $(1 - \frac{k}{n})^m$

D. $1 - (1 - \frac{k}{n})^m$

Solution Answer:(A) Each hash leaves a given bit alone with probability $1 - 1/n$. All km hashings: $(1 - 1/n)^{km}$.

13. Your friend sends you a stream of data items in which none of the elements repeat more than $n/2$ number of times. You run the Misra-Gries heavy-hitter algorithm on this set of items with $k = 1$ (i.e. a single heavy hitter). What do you think is the output of the algorithm?
- (a) Its always the last element of the stream.
 - (b) Its always the second element of the stream.
 - (c) We cannot say which element without knowing the actual stream.
 - (d) It is the center element of the stream, i.e. the element that appears at $(n/2)^{th}$ position in the stream.
 - (e) It is always the first element of the stream

Solution Answer:(C) You can create counter examples for rest of answers.

14. The charging principle of the Misra-Gries algorithm can be generalized from scalar frequency tracking to matrix streams. Suppose rows $a_t \in \mathbb{R}^d$ arrive forming a matrix $A \in \mathbb{R}^{m \times d}$. A sketch $B \in \mathbb{R}^{k \times d}$ is maintained to approximate $A^T A$.

When B is full, the matrix analogue of the Misra-Gries "decrement all counters" step is performed using an SVD-based shrinkage.

Which operation correctly corresponds to this scalar decrement step?

- (A) Subtracting $\sigma_1 I$ from $B^T B$
- (B) Computing SVD $B = U \Sigma V^T$, then setting $\Sigma^2 \leftarrow \Sigma^2 - \sigma_k^2 I$ and reconstructing
- (C) Dropping the first column of B
- (D) Multiplying B by a random matrix R

Solution Answer:(B) In Misra-Gries, a full sketch performs a *global decrement* that reduces all counters uniformly while preserving relative structure.

In the matrix setting, this corresponds to removing uniform energy across all directions.

Using SVD:

$$B = U \Sigma V^T$$

we identify the weakest direction via the smallest singular value σ_k .

We then perform:

$$\Sigma^2 \leftarrow \Sigma^2 - \sigma_k^2 I$$

and reconstruct B from the shrunk spectrum.

This ensures:

$$\|A^T A - B^T B\|_F \text{ remains bounded}$$

Thus, this SVD-based shrinkage is the direct analogue of the Misra-Gries collective decrement step.

15. To support dynamic deletions, each slot of a sketch is replaced with a b -bit counter. Insertions increment k counters and deletions decrement them.

If each counter uses only $b = 4$ bits, which failure mode must be accounted for?

- (A) The filter may suffer false negatives due to counter saturation at $2^4 - 1 = 15$, causing overflow handling issues under deletions.
- (B) The counter array will invert bit values under turnstile updates.
- (C) Load factor is limited to $1/16$ due to bit-width constraints.
- (D) The algorithm requires SVD instead of bitwise operations.

Solution Answer: (A) Each counter is stored using $b = 4$ bits, so the maximum representable value is:

$$2^4 - 1 = 15.$$

Thus, counters can **saturate** at 15. Once saturated: - further increments cannot be represented correctly (overflow or clipping), - deletions may reduce the counter without reflecting true underlying multiplicity.

This breaks the correctness invariant of the sketch and can lead to **false negatives**, where:

true presence of elements is not reflected in the sketch.

Hence, the system must explicitly handle counter saturation and bounded overflow behavior.

16. A stream-processing system uses the Boyer-Moore majority vote algorithm to identify a candidate majority element. For the stream

$$a, b, a, a, b, c, a, d, a, e,$$

which element is reported as the candidate after processing all 10 elements?

- A. a
- B. b
- C. c
- D. No output

Solution Answer: (A) Applying the Boyer-Moore majority vote algorithm to the stream leaves a as the final candidate.

17. Which statement about Misra-Gries $MG(k)$ algorithm is valid?

- A. It always finds all elements occurring more than n/k times.
- B. The output always contains the true heavy hitters if they exist, but may be a superset.
- C. It never misses elements above $n/2$ frequency if $k = 1$.
- D. It cannot work on streams.

Answer: (B) MG finds all heavy hitters, but may also report some elements that are not; a second pass can fix that.

18. An engineer wants to detect all heavy hitters in a stream of length m . The goal is to report all items with frequency:

$$f_x \geq \phi m$$

and exclude all items with:

$$f_x < (\phi - \epsilon)m.$$

Using the Misra-Gries algorithm, what is the optimal choice of k and memory complexity?

- (A) $k = \frac{1}{\phi}$, memory $\mathcal{O}\left(\frac{1}{\phi}(\log n + \log m)\right)$
- (B) $k = \frac{1}{\epsilon}$, memory $\mathcal{O}\left(\frac{1}{\epsilon}(\log n + \log m)\right)$
- (C) $k = \frac{1}{\phi - \epsilon}$, memory $\mathcal{O}\left(\frac{1}{\phi - \epsilon} \cdot n\right)$
- (D) $k = \frac{\phi}{\epsilon^2}$, memory $\mathcal{O}\left(\frac{\phi}{\epsilon^2} \log m\right)$

Solution Answer:(B) Misra-Gries guarantees that the estimation error for any item is at most:

$$\frac{m}{k}.$$

To ensure correct separation between:

$$\phi m \quad \text{and} \quad (\phi - \epsilon)m,$$

the worst-case error must satisfy:

$$\frac{m}{k} \leq \epsilon m.$$

Canceling m :

$$\frac{1}{k} \leq \epsilon \quad \Rightarrow \quad k \geq \frac{1}{\epsilon}.$$

Thus, the optimal choice is:

$$k = \frac{1}{\epsilon}.$$

Each counter stores an item id and a count, requiring $\mathcal{O}(\log n + \log m)$ bits.

Hence total space:

$$\mathcal{O}\left(\frac{1}{\epsilon}(\log n + \log m)\right).$$

19. For the hash family $\mathcal{H} = \{h_1, h_2\}$ shown below,

	1	2	3
h_1	0	1	0
h_2	1	0	1

Consider the pair of keys $(1, 3)$. Based on their collision probability over a uniformly chosen hash function from $\mathcal{H}$, what can you conclude?

- (a) $\mathcal{H}$ is universal.

- (b) $\mathcal{H}$ is not universal.
- (c) Universality cannot be checked using a single pair of keys.

Solution Answer:(B) Since $h_1(1) = h_1(3)$ and $h_2(1) = h_2(3)$, the collision probability is $1 > 1/2$, so $\mathcal{H}$ is **not** universal.

20. A digital library stores millions of documents represented as bag-of-words vectors. A copyright checker needs to answer queries of the form: “Has this exact document appeared in the database before?” The system should use very little memory and answer membership queries in $O(1)$ time, while allowing a very small probability of false positives. Which of the following data structures is most suitable?

- (a) Store all document vectors and perform a linear scan for each query.
- (b) Maintain a Bloom filter over the document identifiers.
- (c) Organize the document vectors in a k -d tree.

Solution Answer:(B) Since only approximate membership queries with a very low false positive rate are required in $O(1)$ time and with low memory usage.